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COURSE CODE: SBT-DF203-BASIC NETWORKING SKILLS

LAB NAME: LAB4 SMTP EMAIL TRAFFIC FORENSICS

DATE: 15/09/26

INSTRUCTOR: AMINU IDRIS

International Cybersecurity and Digital Forensics Academy (ICDFA)

SBT-DF203: Basic Networking Skills for Digital Forensics

LAB 4: SMTP Email Traffic Forensics

Three-Week Delivery Block 2/3 | 12-18 September 2026

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

Important Pre-Lab Instruction

Read the matching lecture slide deck before executing the practical. Work only inside the ICDFA-approved lab or your own isolated virtual environment. Record every command, observation and screenshot as contemporaneous evidence.

Course Code	SBT-DF203
Course Title	Basic Networking Skills for Digital Forensics
Lab Number	Lab 4
Lab Title	SMTP Email Traffic Forensics
Delivery Block	2/3 of 3
Scheduled Dates	12-18 September 2026
Estimated Duration	3-4 hours practical work plus report writing
Mode	Individual practical lab in an authorized virtual environment
Required Evidence	smtp.pcap supplied sample capture

1. Source Materials and Slide Alignment

- Email Traffic Forensics: SMTP purpose, ports 25/587/465, plaintext commands, STARTTLS/SMTPS distinctions, Base64 encoding, stream reassembly and email-content analysis.

- The slide practical uses smtp.pcap containing Microsoft Outlook traffic and requires timestamps, client, content, ports, IP addresses and MAC addresses.
- The exercise treats credentials as sensitive evidence and requires protected reporting.

Reading Requirement Before Lab Execution

The lab deliberately follows the sequence, terminology and core exercises presented in the uploaded SBT-DF203 slides. Commands have been corrected where needed for Linux syntax and forensic repeatability.

2. Lab Scenario

A historical SMTP packet capture has been supplied for analysis. You must determine when the email exchange occurred, identify the client and server, reconstruct SMTP commands and response codes, decode relevant Base64 authentication fields, recover the message content, and document the network path without exposing sensitive data unnecessarily.

3. Learning Outcomes

- Explain the role of SMTP and distinguish plaintext SMTP, STARTTLS and implicit TLS.
- Identify SMTP commands, server response codes and authentication exchanges.
- Reassemble an SMTP TCP stream and reconstruct message headers and body.
- Decode Base64 values in a controlled offline manner.
- Extract timestamps, client software, IP addresses, ports and MAC addresses.
- Assess evidential limitations when encryption is used or capture is incomplete.

4. Legal, Ethical and Safety Rules

- Use only systems, packet captures and networks for which you have explicit authorization.
- Keep the lab isolated from production, campus, office and public networks. Use NAT or host-only virtual networking as instructed.
- Do not collect, disclose or reuse real credentials, personal messages or confidential traffic.
- Preserve original capture files. Perform analysis on verified working copies and record SHA-256 hashes.
- Stop any traffic-generation or interception process immediately after the required evidence is captured.
- Restore firewall, ARP, forwarding and network settings before closing the lab.
- Treat any recovered username, password or message content as confidential assessment evidence. Mask sensitive values in screenshots unless the instructor explicitly requires the full value.

5. Required Tools and Environment

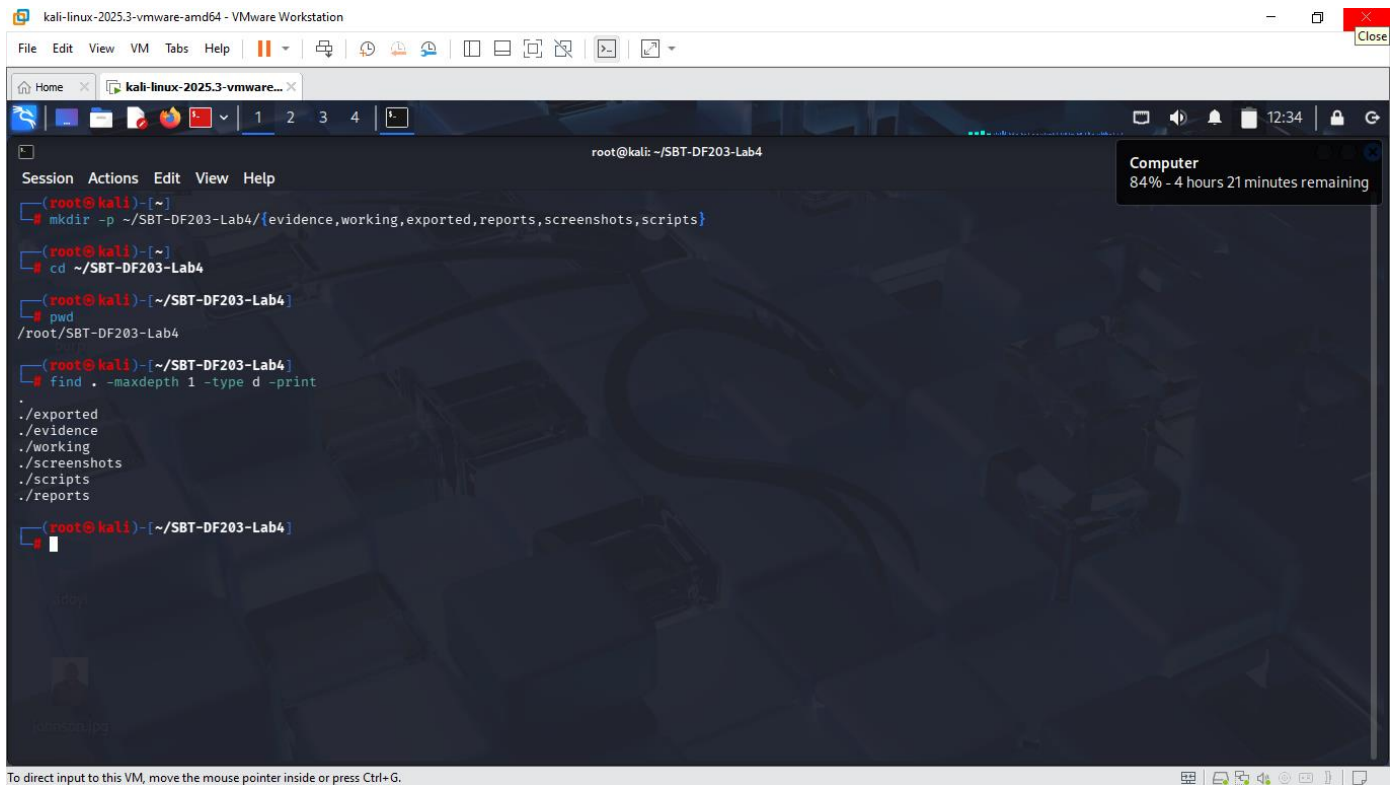
Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu VM	Analyze the capture.
Packet tools	Wireshark and tshark	Filter and reconstruct SMTP streams.
Decoder	Python 3 base64 module	Decode Base64 strings offline.
Hashing	sha256sum	Preserve evidence integrity.
Evidence	smtp.pcap	Historical training traffic.
Text editor	Any local editor	Document recovered message safely.

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab4
pwd
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

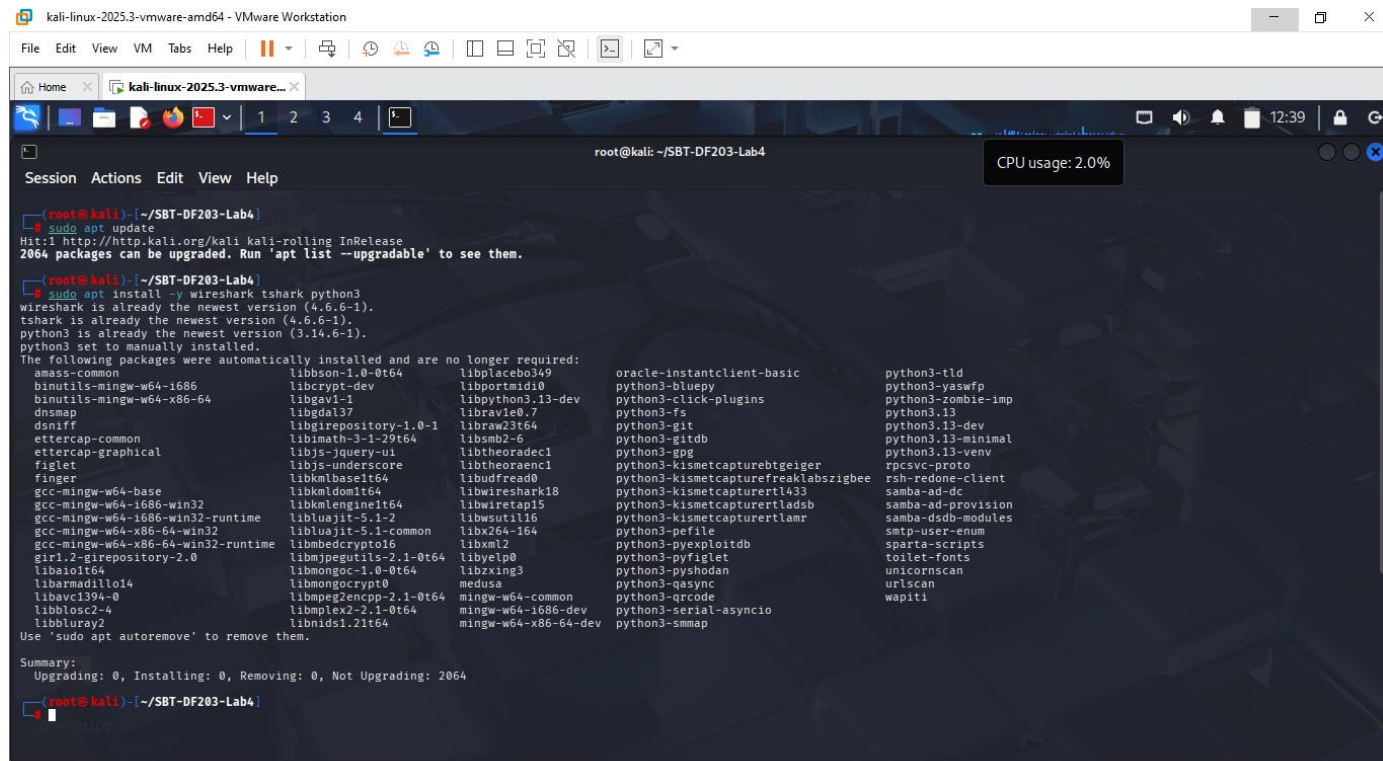
ANSWER:



```
sudo apt update
sudo apt install -y wireshark tshark python3
wget -O evidence/smtp.pcap
'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap'
cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap
sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
```

Screenshot required: smtp.pcap file details, capinfos summary and matching original/working-copy hashes.

ANSWER:



```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
root@kali: ~/SBT-DF203-Lab4
CPU usage: 2.0%

Session Actions Edit View Help

(root@kali) ~/SBT-DF203-Lab4
# sudo apt update
Hit:1 http://http.kali.org/kali kali-rolling InRelease
2064 packages can be upgraded. Run 'apt list --upgradable' to see them.

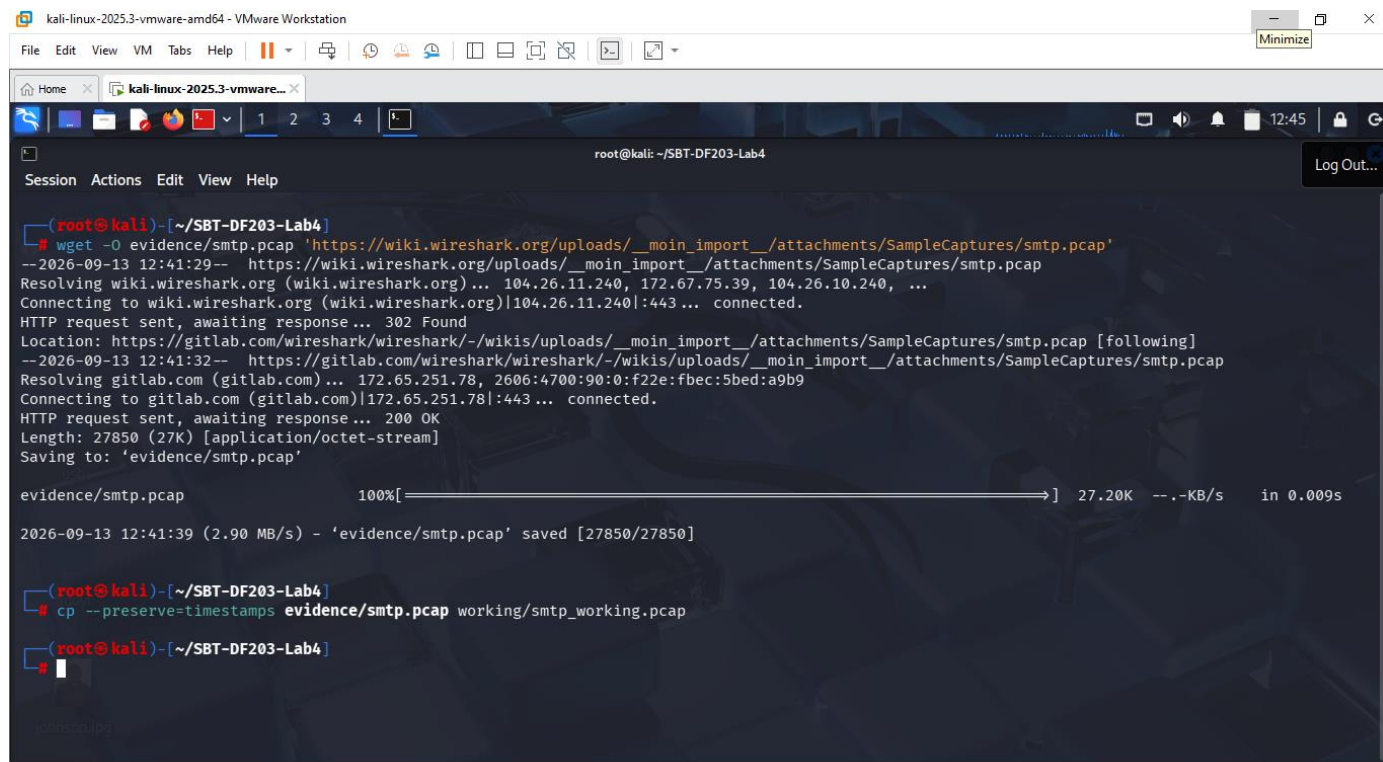
(root@kali) ~/SBT-DF203-Lab4
# sudo apt install -y wireshark tshark python3
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
python3 is already the newest version (3.14.6-1).
python3 set to manually installed.
The following packages were automatically installed and are no longer required:
amass-common libbson-1.0-0t64 libplacebo349 python3-tld
binutils-mingw-w64-x86_64 libcrypt-dev libportmidi0 python3-bluepy
binutils-mingw-w64-x86_64 libgavi1-1 libpython3.13-dev python3-click-plugins
dnsmasq libgdal37 libbravie0.7 python3-fs
dsniff libgirepository-1.0-1 libraw23t64 python3-git python3-zombie-imp
ettercap-common libimath-3-1-29t64 libsmB2-6 python3-gitdb python3.13-dev
ettercap-graphical libjs-jquery-ui libtheoradec1 python3-gpg python3.13-minimal
figlet libjs-underscore libtheoraec1 python3-kismetcapturebtgeiger python3.13-venv
finger libkmlbase1t64 libudfread0 python3-kismetcapturefreaklabszigbee rpcsvc-proto
gcc-mingw-w64-base libkmlidom1t64 libwireshark18 python3-kismetcapturefreaklabszigbee rsh-redone-client
gcc-mingw-w64-1686-win32 libkmlengine1t64 libwireshark15 python3-kismetcapturefreaklabszigbee samba-ad-dc
gcc-mingw-w64-1686-win32-runtime liblua5.1-2 libwsutil16 python3-kismetcapturefreaklabszigbee samba-ad-provision
gcc-mingw-w64-x86_64-win32 liblua5.1-common libx264-164 python3-kismetcapturefreaklabszigbee samba-dsdb-modules
gcc-mingw-w64-x86_64-win32-runtime libmbdcrpt016 libxml2 python3-kismetcapturefreaklabszigbee samba-dsdb-modules
girl2-girepository-2.0 libmjpegutils-2.1-0t64 libyelp0 python3-pyfiglet smtp-user-enum
libaio1t64 libmongoc-1.0-0t64 libzxing3 python3-pyshodan sparta-scripts
libarmadillo14 libmonocrypt0 medusa python3-qasync toilet-fonts
libavc1394-0 libmpeg2encpp-2.1-0t64 mingw-w64-common python3-qrcode unicornsans
libblosc2-4 libmpeg2-2-1-0t64 mingw-w64-1686-dev python3-qrcode unicornsans
libbluray2 libnids1.21t64 mingw-w64-x86_64-dev python3-smmap wapiti

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2064

(root@kali) ~/SBT-DF203-Lab4

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.



```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
root@kali: ~/SBT-DF203-Lab4
Log Out...

Session Actions Edit View Help

(root@kali) ~/SBT-DF203-Lab4
# wget -O evidence/smtp.pcap 'https://wiki.wireshark.org/uploads/_moin_import_/attachments/SampleCaptures/smtp.pcap'
--2026-09-13 12:41:29-- https://wiki.wireshark.org/uploads/_moin_import_/attachments/SampleCaptures/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org)... 104.26.11.240, 172.67.75.39, 104.26.10.240, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.11.240|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/_moin_import_/attachments/SampleCaptures/smtp.pcap [following]
--2026-09-13 12:41:32-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/_moin_import_/attachments/SampleCaptures/smtp.pcap
Resolving gitlab.com (gitlab.com)... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/smtp.pcap'

evidence/smtp.pcap 100%[=====] 27.20K --.-KB/s in 0.009s

2026-09-13 12:41:39 (2.90 MB/s) - 'evidence/smtp.pcap' saved [27850/27850]

(root@kali) ~/SBT-DF203-Lab4
# cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap

(root@kali) ~/SBT-DF203-Lab4

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Home kali-linux-2025.3-vmware...x
1 2 3 4
root@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
# sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/smtp_working.pcap

# capinfos evidence/smtp.pcap | tee reports/smtp_capinfos.txt
File name: evidence/smtp.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 60
File size: 27 kB
Data size: 26 kB
Capture duration: 9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060
Latest packet time: 2009-10-05 02:06:16.690444
Data byte rate: 2,920 bytes/s
Data bit rate: 23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256: 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1: 3def1a1fed849e7f23b66e6925dfff05845a400b
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
  Encapsulation = Ethernet (1 - ether)
  Capture length = 65535
  Time precision = microseconds (6)
  Time ticks per second = 1000000
  Number of stat entries = 0
  Number of packets = 60

root@kali: ~/SBT-DF203-Lab4
Computer
81% - 4 hours 44 minutes remaining

```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab4-Adoyi Nathaniel Johnson
Trainee name	ADOYI NATHANIEL JOHNSON
Date and time started	13/09/26 05:15PM
Evidence file name(s)	SBT-DF203_LAB4_SMTP EMAIL TRAFFIC FORENSICS
Source or generation method	ROOT@KALI LINUX COMMAND
Original SHA-256	sha256sum evidence/smtp.pcap working/smtp_working.pcap tee reports/smtp_capture_hashes.txt
Working-copy SHA-256	17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/smtp.pcap 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/smtp_working.pcap
Analysis workstation/VM	ROOT@KALI
Notes on any changes	SMTP and PCAP are Working

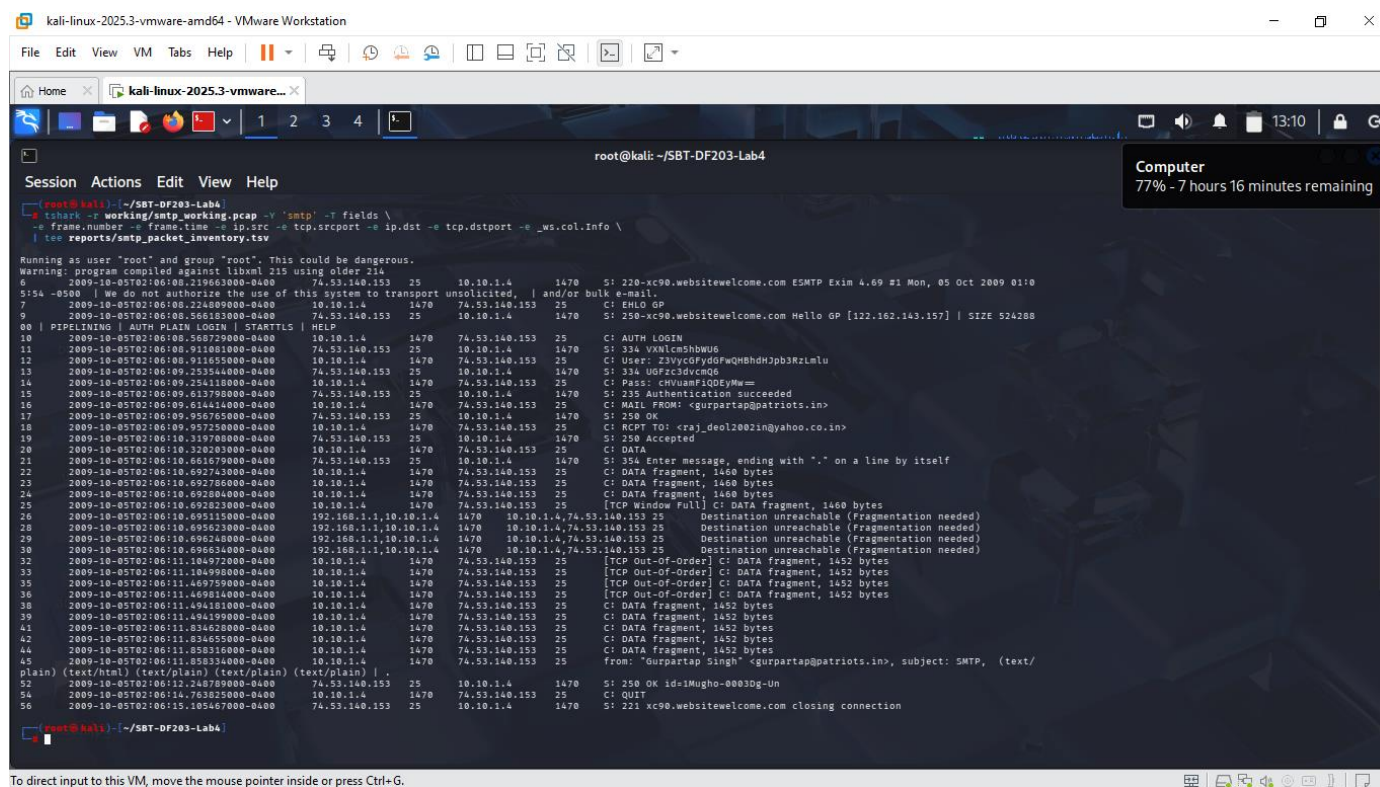
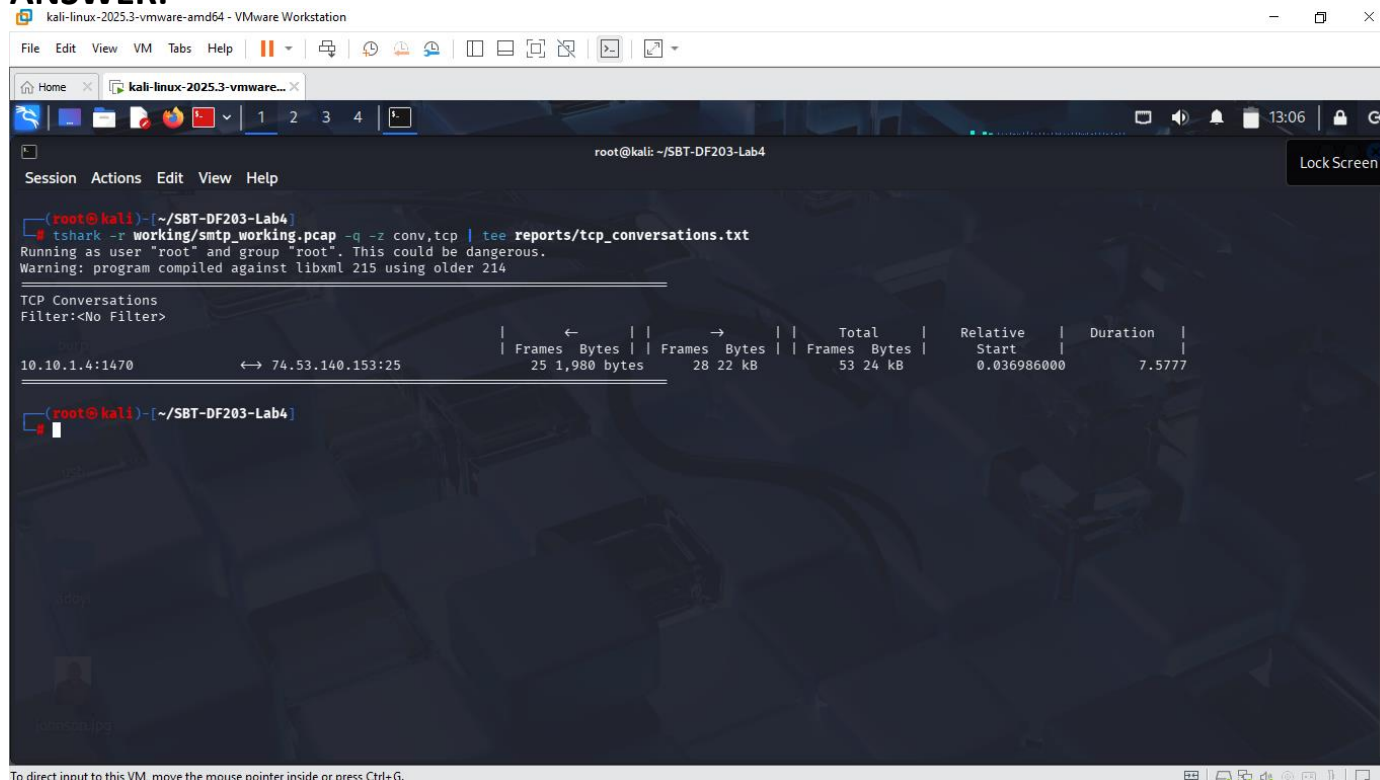
8. Part A - Inventory the Capture and Locate SMTP Streams

```
tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
```

```
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.Info \
| tee reports/smtp_packet_inventory.tsv
```

Record the SMTP server port, client ephemeral port, first packet time, last packet time and number of SMTP frames.

ANSWER:



9. Part B - Identify Commands and Response Codes

Sequence	Direction	Command/Code	Meaning	Timestamp
1	Server to client	220	Service ready	7t2009-10-05T02:06:08.224809000-0400t10.10.1.4t74.53.140.153tEHLOtGPtt
2	Client to server	EHLO/HELO	Client greeting	7t2009-10-05T02:06:08.224809000-0400t10.10.1.4t74.53.140.153tEHLOtGPtt
3	Client to server	AUTH	Authentication method	10t2009-10-05T02:06:08.568729000-0400t10.10.1.4t74.53.140.153tAUTHtLOGINtt
4	Client to server	MAIL FROM	Envelope sender	16t2009-10-05T02:06:09.614414000-0400t10.10.1.4t74.53.140.153tMAILtFROM:<gurpar\tap@paltrio\ts.in>tt
5	Client to server	RCPT TO	Envelope recipient	18t2009-10-05T02:06:09.957250000-0400t10.10.1.4t74.53.140.153tRCPTtTO:<raj_deol2002in@yahoo.co.in>tt
6	Client to server	DATA	Message content begins	20t2009-10-05T02:06:10.320203000-0400t10.10.1.4t74.53.140.153tDATAttt
7	Client to server	QUIT	Session termination	54t2009-10-05T02:06:14.763825000-0400t10.10.1.4t74.53.140.153tQUITttt

```
tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
  -e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
  -e smtp.response.code -e smtp.response.parameter \
  | tee reports/smtp_commands_responses.tsv
```

ANSWER:

```

root@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help

root@kali:~/SBT-DF203-Lab4
# tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code -e smtp.rsp.parameter \
-E header=y -E separator=t \
| tee reports/smtp_commands_responses.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
frame.numberframe.timeframe.timeip.srcip.dstsmtp.req.commandsmtp.req.parametersmtp.response.codetsmtp.rsp.parameter
6t2009-10-05T02:06:08.219663000-0400t74.53.140.153t10.10.1.4ttt220txc90.websi\tewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oc\t 2009 01:05:54 -0500 ,We do no\t au\thorize
\tthe use of \this sys\tem \to \transport unsolici\ted, ,and/or bulk e-mail.
7t2009-10-05T02:06:08.224809000-0400t10.10.1.4t74.53.140.153tEHLOtGPtt
9t2009-10-05T02:06:08.566183000-0400t74.53.140.153t10.10.1.4ttt250txc90.websi\tewelcome.com Hello GP [122.162.143.157],SIZE 52428800,PIPELINING,AUTH PLAIN LOGIN,START
TLS,HELP
10t2009-10-05T02:06:08.568729000-0400t10.10.1.4t74.53.140.153tAUTHtLOGIntt
11t2009-10-05T02:06:08.911081000-0400t74.53.140.153t10.10.1.4ttt334tVXNlcm5hbWU6
12t2009-10-05T02:06:08.911655000-0400t10.10.1.4t74.53.140.153t10.10.1.4ttt334tUGFzc3dvcnQ6
13t2009-10-05T02:06:09.253544000-0400t74.53.140.153t10.10.1.4ttt334tUGFzc3dvcnQ6
14t2009-10-05T02:06:09.254118000-0400t10.10.1.4t74.53.140.153t10.10.1.4ttt334tUGFzc3dvcnQ6
15t2009-10-05T02:06:09.613798000-0400t74.53.140.153t10.10.1.4ttt235tAu\then\tica\tion succeeded
16t2009-10-05T02:06:09.614414000-0400t10.10.1.4t74.53.140.153tMAILtFROM: <gurpar\tap@pa\trio\ts.in>t
17t2009-10-05T02:06:09.956765000-0400t74.53.140.153t10.10.1.4ttt250tOK
18t2009-10-05T02:06:09.957250000-0400t10.10.1.4t74.53.140.153tRCPTtTO: <raj_deol2002in@yahoo.co.in>t
19t2009-10-05T02:06:10.319708000-0400t74.53.140.153t10.10.1.4ttt250tAccep\ted
20t2009-10-05T02:06:10.320203000-0400t10.10.1.4t74.53.140.153tDATAttt
21t2009-10-05T02:06:10.661679000-0400t74.53.140.153t10.10.1.4ttt354tEn\ter message, ending with "." on a line by i\tself
52t2009-10-05T02:06:12.248789000-0400t74.53.140.153t10.10.1.4ttt250tOK id=1Mugho-0003Dg-Un
54t2009-10-05T02:06:14.763825000-0400t10.10.1.4t74.53.140.153tQUITttt
56t2009-10-05T02:06:15.105467000-0400t74.53.140.153t10.10.1.4ttt221txc90.websi\tewelcome.com closing connec\tion

```

10. Part C - Decode Base64 Authentication Evidence

Base64 is encoding, not encryption. Copy only the required encoded values into the offline script. Do not use online decoders.

```

python3 - <<'PYLAB'
import base64
samples = {
    'username': 'REPLACE_WITH_CAPTURED_BASE64',
    'password': 'REPLACE_WITH_CAPTURED_BASE64'
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB

```

Reporting Rule

In the public-facing report body, show a masked value such as analyst@example.com and P*****3. Place full decoded values only in the protected evidence appendix if required by the instructor.

ANSWER:

```

kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
(root@kali)~[~/SBT-DF203-Lab4]
python3 - <<'PYLAB'
import base64
samples = {
    'username': 'Z3VycGFydGFwQH8hdHJpb3RzLmlu',
    'password': 'cHVuamFiQDEyMw=='
}
for label, value in samples.items():
    try:
        decoded = base64.b64decode(value).decode('utf-8', errors='replace')
        print(f'{label}: {decoded}')
    except Exception as exc:
        print(f'{label}: decode failed: {exc}')
PYLAB
username: gurpartap@patriots.in
password: punjab@123
(root@kali)~[~/SBT-DF203-Lab4]

```

username: gurpartap@patriots.in

password: punjab@123

11. Part D - Reconstruct the Email Message

1. In Wireshark select an SMTP packet and choose Follow > TCP Stream.
2. Identify SMTP envelope fields separately from RFC message headers.
3. Record Date, From, To, Subject, Message-ID, MIME-Version, Content-Type and User-Agent/X-Mailer where present.
4. Determine whether the body is plain text, HTML, multipart or contains an attachment.
5. Save a redacted text reconstruction to reports/reconstructed_email_redacted.txt.

```

# Replace STREAM with the relevant stream number
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,STREAM \
| tee reports/smtp_stream_STREAM.txt

```

```

grep -Ei '^(Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer):' \
reports/smtp_stream_STREAM.txt | tee reports/message_headers.txt

```

ANSWER:

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64...
root@kali: ~/SBT-DF203-Lab4
Session Actions Edit View Help
(root@kali) ~/SBT-DF203-Lab4
tshark -r working/smtp_working.pcap -q -z follow,tcp,ascii,0 \
| tee reports/smtp_stream_0.txt
grep -Ei '^([Date|From|To|Subject|Message-ID|MIME-Version|Content-Type|User-Agent|X-Mailer]):' \
reports/smtp_stream_0.txt | tee reports/message_headers.txt
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
Follow: tcp,ascii
Filter: tcp.stream eq 0
Node 0: 10.10.1.4:1470
Node 1: 74.53.140.153:25
181
220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500
220-We do not authorize the use of this system to transport unsolicited,
220 and/or bulk e-mail.
9
EHLO GP
137
250-xc90.websitewelcome.com Hello GP [122.162.143.157]
250-SIZE 52428800
250-PIPELINING
250-AUTH PLAIN LOGIN
250-STARTTLS
250 HELP
12
* new update/packages checker (vUpdate)
* Lots of bugfixes
-----_NextPart_000_00
29
04_01CA45B0.095693F0--
.
28
250 OK id=1Mugho-0003Dg-Un
6
QUIT
48
221 xc90.websitewelcome.com closing connection
From: "Gurpartap Singh" <gurpartap@patriots.in>
To: <raj_deol2002in@yahoo.co.in>
Subject: SMTP
Date: Mon, 5 Oct 2009 11:36:07 +0530
Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>
MIME-Version: 1.0
Content-Type: multipart/mixed;
X-Mailer: Microsoft Office Outlook 12.0
Content-Type: multipart/alternative;
Content-Type: text/plain;
Content-Type: text/html;
Content-Type: text/plain;
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

12. Part E - Determine Client, Hosts and Network Metadata

```
tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
-e ip.dst -e tcp.dstport -e tcp.stream \
| tee reports/smtp_network_metadata.tsv

tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
| tee reports/client_indicators.tsv
```

ANSWER:

The screenshot shows a Kali Linux terminal window with the following commands and output:

```
root@kali:~/SBT-DF203-Lab4
Session Actions Edit View Help
-e ip.dst -e tcp.dstport -e tcp.stream \
| tee reports/smtp_network_metadata.tsv
tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
| tee reports/client_indicators.tsv
```

Warning: program compiled against libssl 215 using older 214

Frame Number	Time Epoch	Eth Src	Eth Dst	IP Src	IP Dst	TCP Src Port	TCP Dst Port	TCP Stream
6	125422768.219680000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
7	125422768.224800000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
8	125422768.246100000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
10	125422768.508790000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
11	125422768.913810000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
12	125422768.916550000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
13	125422768.235440000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
14	125422768.254110000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
15	125422768.613700000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
16	125422768.634610000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
17	125422768.956760000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
18	125422768.957200000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
19	125422770.319700000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
20	125422770.320200000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
21	125422770.601670000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
22	125422770.602740000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
23	125422770.607800000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
24	125422770.692800000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
25	125422770.692820000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
26	125422770.695110000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	192.168.1.1,10.10.1.4	10.10.1.4,74.53.140.153	25	10.10.1.4	1470 0
27	125422770.696240000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
29	125422770.696240000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	192.168.1.1,10.10.1.4	10.10.1.4,74.53.140.153	25	10.10.1.4	1470 0
30	125422770.696240000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
31	125422771.180970000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
33	125422771.184990000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
35	125422771.184970000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
36	125422771.440610000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
38	125422771.440610000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
39	125422771.440610000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
41	125422771.831310000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
42	125422771.831310000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
44	125422771.831310000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
45	125422771.831310000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
46	125422771.831310000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
48	125422771.248790000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0
50	125422771.703250000	00:00:1c:3c:17:c2	00:1f:33:09:01:60	10.10.1.4	74.53.140.153	25	74.53.140.153	25 0
56	125422771.105467000	00:1f:33:09:01:60	00:00:1c:3c:17:c2	74.53.140.153	10.10.1.4	25	10.10.1.4	1470 0

Warning: program compiled against libssl 215 using older 214

```
0 Line-based text data (47 lines)
26 0 Line-based text data (12 lines)
```

13. Part F - Encryption and Evidential Limitations

- Check whether STARTTLS is requested and whether subsequent payload becomes TLS.
- If traffic is encrypted, document metadata that remains visible: endpoints, ports, timing, certificate/session information and byte counts.
- Do not claim message content or credentials when the capture does not reveal them.
- Explain that port number alone does not prove encryption; protocol negotiation and packet dissection must be examined.

14. Required Findings Worksheet

Question	Finding
When did the SMTP session start and end?	START: 5 Oct 2009, 06:06:07.529 UTC. END: 5 Oct 2009, 06:06:15.107 UTC.
Client IP/MAC and port	IP: 10.10.1.4; MAC: 00:1f:33:d9:81:60; Port: 1470
Server IP/MAC and port	
SMTP server banner	220-xc90.websitewelcome.com
Client software	Microsoft Office Outlook 12.0
Authentication method	AUTH LOGIN
Envelope sender and recipient	AN EMAIL FROM: gurpartap@patriots.in
Message From/To/Subject	TO: raj_deol2002in@yahoo.co.in
Message body type	MIME multipart/mixed, containing text/plain and text/html parts
Attachment present?	YES, Plaintext named tes.txt
STARTTLS/TLS observed?	The SMTP exchange did NOT show STARTTLS, and the authentication/data were transmitted without TLS encryption.
Key limitations	The PCAP only shows network activity captured at that point. It does not independently prove the real-world identity of the sender, and email headers can be spoofed. The capture also contains authentication information that should be treated as sensitive.

Required Screenshots Checklist

- ☐ Capture hashes and capinfos summary.
- ☐ SMTP conversation list.
- ☐ 220 service-ready response.
- ☐ EHLO/HELO and AUTH exchange.
- ☐ Masked Base64 decoding result.
- ☐ MAIL FROM, RCPT TO and DATA.
- ☐ Follow TCP Stream reconstruction.
- ☐ Email headers and client software.
- ☐ Source/destination IP, ports and MAC addresses.
- ☐ TLS/STARTTLS assessment.

Student Report Template

6. Executive summary.
7. Evidence integrity and scope.
8. SMTP timeline.
9. Command/response analysis.
10. Authentication evidence and redaction method.
11. Email reconstruction.
12. Client/server and network metadata.
13. Encryption assessment and limitations.
14. Conclusions and appendix.

Submission Requirements

- One PDF report named SBT-DF203-Lab4_RegNo_FullName.pdf.
- The original or generated PCAP/PCAPNG evidence file and its SHA-256 hash text file.
- A reports folder containing exported CSV/TXT command output and any recovered objects.
- Screenshots must be readable, numbered and referenced in the report narrative.
- Compress the report and evidence outputs into one ZIP named with your registration number and lab number.
- Do not submit passwords or decoded credentials in public discussion channels; include them only in the protected assessment submission where required.

Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Evidence handling	10	Original preserved, working copy and hashes documented.
SMTP command analysis	20	Correct commands, codes, direction and timeline.
Authentication analysis	15	Correct offline decoding and safe redaction.
Message reconstruction	25	Accurate headers, body type, client and content findings.
Network metadata and encryption	20	Correct endpoints, ports, MACs and TLS assessment.
Report quality	10	Professional, concise and evidence-linked report.

Instructor Quick Answer Guide

Checkpoint	Expected Observation
Service-ready code	220.
Client greeting	HELO or EHLO.
Authentication evidence	Often AUTH LOGIN with Base64-encoded fields in plaintext SMTP.
Message data	Begins after DATA and ends with SMTP termination sequence.
Client indicator	Slide capture identifies Microsoft Outlook 12.0.
Important limitation	Base64 is reversible encoding; TLS would hide application content.

Command Reference Summary

Command	Purpose
smtp	Wireshark display filter for SMTP.
smtp.req / smtp.rsp	Filter SMTP requests and responses.
Follow TCP Stream	Reconstruct the application conversation.
base64.b64decode	Decode Base64 offline.
capinfos file	Summarize capture metadata.
eth.addr / ip.addr / tcp.port	Endpoint filters.

Academic Integrity and Authorization Statement

Student Declaration

I confirm that I completed this lab in an authorized environment, preserved the supplied evidence, did not target any third-party system, and accurately documented my own commands, observations and conclusions.